

30 April 2026

Editor-in-Chief
Proceedings of the Institution of Mechanical Engineers,
Part D: Journal of Automobile Engineering
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Dear Editor,

We submit for your consideration the manuscript entitled “BowTie Risk Analysis of the Valve Train in a High-Performance VW AP 1.8 Cylinder Head Operating at 10,000 rpm” as an Original Article.

The manuscript proposes a quantitative-coupled BowTie procedure for design-stage reliability assessment of the valve train of a high-performance spark-ignition cylinder head, and applies it to a top event of catastrophic significance for high-rpm engines: the loss of follower contact between valve and cam at 10,000 rpm. Beyond the qualitative bow diagram, each preventive barrier is associated with a deterministic indicator computed from the analytical sizing of the intake valve and the dual chrome-silicon valve springs (dynamic follower-contact ratio, natural-frequency margin against cam harmonics, coil-bind clearance, and Goodman fatigue factor of safety), with literature-anchored pass criteria. The procedure is qualitatively validated against four documented valve-train and engine-component failure cases from the recent literature, and a concrete experimental validation roadmap — spring-rig modal verification, motored-engine speed sweep, and inner-spring fatigue campaign, with explicit acceptance criteria — is specified for empirical confirmation. The case study is the Volkswagen AP 1.8 platform — a base engine with a wide installed base in the Brazilian automotive market — for which a high-performance cylinder head is currently being developed within a Master’s research project at the Universidade Católica de Petrópolis.

We believe *Proceedings of the IMechE, Part D: Journal of Automobile Engineering* is the most appropriate venue because (i) the journal has a long publication history of design and reliability analyses of internal-combustion-engine subsystems, including valve-train and cam-train dynamics; (ii) the proposed quantitative-coupling layer addresses a recurring gap between qualitative risk diagrams and engineering design verification, which is of direct interest to the journal’s automotive-engineering readership; and (iii) the application domain — valve-train dynamics at extreme rotational speed in a passenger-car-derived platform — aligns with the journal’s traditional scope.

The work is co-authored by Darlan Costa Porto (corresponding author, Master’s student in the Engineering Systems programme), Heitor Oliveira Gonçalves (faculty of the same programme, ORCID 0000-0002-5866-2213), Florian Pradelle (PUC-Rio collaborator), and José Cristiano Pereira (advisor of the underlying dissertation). Author contributions are detailed in the CRediT statement at the end of the manuscript.

The accompanying manuscript PDF was prepared in a generic LaTeX double-column template. The LaTeX source can be ported to the SAGE journal style file (`sagej.cls`) on acceptance, in line with the journal’s usual production workflow. The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All authors have approved the submission, and the authors declare no competing financial interests.

We respectfully suggest the following independent reviewers as potentially appropriate. None has co-authored work cited in this manuscript with the present authors, and none is affiliated with the Universidade Católica de Petrópolis. Institutional contact details can be located on each researcher’s institutional page.

1. Prof. Chiara Soffritti — Department of Engineering, University of Ferrara, Italy. Co-author of a 2018 paper on valve-train wear in four-cylinder diesel engines (DOI 10.1016/j.engfailanal.2018.06.022), directly comparable to the present case study.
2. Dr. José Antonio Vélez Godiño — Departamento de Ingeniería Energética, Universidad de Sevilla, Spain. First author of a 2019 paper on overhead valve-train failure in urban buses (DOI 10.1016/j.engfailanal.2018.11.003); competent in field-data interpretation of valve-train degradation.
3. Prof. Faisal Khan — Mary Kay O'Connor Process Safety Center, Texas A&M University, USA. Co-author of the BowTie–Bayesian-network mapping (Khakzad, Khan and Amyotte, 2013, DOI 10.1016/j.psep.2012.01.005); senior authority on barrier-based risk methods.
4. Prof. Frank Guldenmund — Section Safety and Security Science, TU Delft, Netherlands. Co-author of the canonical BowTie review (de Ruijter and Guldenmund, 2016, DOI 10.1016/j.ssci.2016.03.001).
5. Dr. Jonas Aust — Department of Mechanical Engineering, University of Canterbury, New Zealand. First author of the BowTie methodology paper for aircraft-engine borescope inspection (*Aerospace*, 2019, DOI 10.3390/aerospace6100110); experienced in BowTie applied to engine maintenance.

The final selection of reviewers is, of course, at the editor's discretion.

We look forward to your editorial decision and remain available for any clarification.

Yours sincerely,

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